METHODOLOGY DOCUMENT
MSCI SUSTAINABILITY AND CLIMATE

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# MSCI ESG Ratings Methodology: Product Carbon Footprint Key Issue

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# Introduction

This document provides essential information on the components of the Product Carbon Footprint Key Issue to enable users of ESG Ratings to understand how MSCI Sustainability & Climate's (MSCI's),[1] outputs are determined. Product Carbon Footprint is a Key Issue in the Environmental Pillar of the MSCI ESG Ratings model. Companies are evaluated on the carbon intensity of their products and their ability to reduce the carbon footprint in their supply chains or in the use of their products and services.

For additional details on the MSCI ESG Ratings Model, refer to the Data, ratings and scores section of "ESG Ratings Methodology."

**Exhibit 1: MSCI ESG Key Issue hierarchy**

| 3 Pillars | 10 Themes | 33 ESG Key Issues |
|---|---|---|
| **Environmental** | **Climate Change** | Carbon Emissions |
| | | Climate Change Vulnerability |
| | | Financing Environmental Impact |
| | | **Product Carbon Footprint** |
| | Natural Capital | Biodiversity & Land Use |
| | | Raw Material Sourcing |
| | | Water Stress |
| | Pollution & Waste | Electronic Waste |
| | | Packaging Material & Waste |
| | | Toxic Emissions & Waste |
| | Environmental Opportunities | Opportunities in Clean Tech |
| | | Opportunities in Green Building |
| | | Opportunities in Renewable Energy |
| **Social** | Human Capital | Health & Safety |
| | | Human Capital Development |
| | | Labor Management |
| | | Supply Chain Labor Standards |
| | Product Liability | Chemical Safety |
| | | Consumer Financial Protection |
| | | Privacy & Data Security |
| | | Product Safety & Quality |
| | | Responsible Investment |
| | Stakeholder Opposition | Community Relations |
| | | Controversial Sourcing |
| | Social Opportunities | Access to Finance |
| | | Access to Health Care |
| | | Opportunities in Nutrition & Health |

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| 3 Pillars | 10 Themes | 33 ESG Key Issues |
|---|---|---|
| **Governance** | Corporate Governance | Board |
| | | Pay |
| | | Ownership & Control |
| | | Accounting |
| | Corporate Behavior | Business Ethics |
| | | Tax Transparency |

## Risks associated with this Key Issue

- Increased costs from higher and more volatile energy prices upstream in the value chain (affecting raw materials, input, and distribution costs).
- Regulatory risks associated with downstream emissions, e.g., penalties resulting from vehicle fleet emissions standards in the automobile industry.

## Product Carbon Footprint Key Issue score

The Product Carbon Footprint Key Issue score evaluates the company's level of exposure to, and management of, risks on this Key Issue. The Key Issue score on this Key Issue is based on the Exposure score and the Management score using the formula below. The Exposure score and the Management score are combined such that a higher level of exposure requires a higher level of demonstrated management capability in order to achieve the same overall Key Issue score.

$$KI_i = 7 - (\max(EXP_i, 2) - MGMT_i)$$

Where:

- $KI_i$ is the Key Issue score for company $i$.
- $EXP_i$ is the Exposure score for company $i$.
- $MGMT_i$ is the Management score for company $i$.

## Product Carbon Footprint Management framework

The table below outlines the indicators mapped to their respective components that inform the Product Carbon Footprint Management score, excluding controversies. MSCI S&C assigns each indicator a default weight. Indicator weights may be adjusted at the company level to reflect the relevance of the indicator to the company's business activities. If a company has no exposure to the business activities covered by an indicator, the indicator is assigned a weight of zero.

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| Component | Sustainability Indicator |
|---|---|
| Product Carbon Footprint Practices score | Assessment of energy consumption and/or carbon emissions of suppliers |
| | Extent of product carbon footprint assessments |
| | Targets to reduce Scope 3 upstream carbon emissions |
| | Energy consumption or carbon emissions assessment of distribution and store operations |
| | Energy consumption or carbon emissions assessment of manufacturing operations (including suppliers) |
| | Energy consumption or carbon emissions assessment of raw materials sourcing |
| | Energy consumption or carbon emissions assessment of transportation and logistics |
| | Extent of energy consumption or carbon emissions reduction programs in distribution centers or store operations |
| | Extent of energy consumption or carbon emissions reduction programs in manufacturing operations (including suppliers) |
| | Extent of energy consumption or carbon emissions reduction programs in raw materials sourcing |
| | Extent of energy consumption or carbon emissions reduction programs in transportation and logistics |

## Product Carbon Footprint Management score

The Product Carbon Footprint Management score evaluates the company's ability to manage its exposure to risks on this Key Issue. It is based on weighted average of the score associated with each relevant indicators, modified by controversies.

This score is derived from indicators that are scored on a 0-10 scale, with 10 corresponding to best practice. Refer to the Analyzing risk management section of "ESG Ratings Methodology" for additional details. Sources are company disclosures except where otherwise indicated.

The following formula is used to calculate the Product Carbon Footprint Management score - Excluding Controversies:

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$$MGMT_i = \sum Indicator_Score_{j,i} * Indicator_Weight_{j,i}$$

Where:

- $MGMT_i$ is the Management score - Excluding Controversies for company $i$.
- $Indicator_Score_{j,i}$ is the score of indicator $j$ for company $i$.
- $Indicator_Weight_{j,i}$ is the weight of indicator $j$ for company $i$.

The following formula is used to calculate the scores for the components

$$COMP_{c,i} = \sum_{j \in c} Indicator_Score_{j,i} * Indicator_Weight_Scaled_{j,i}$$

Where:

- $COMP_{c,i}$ is the component score for component $c$, for company $i$.
- $Indicator_Score_{j,i}$ is the score of indicator $j$ for company $i$.
- $Indicator_Weight_Scaled_{j,i}$ is the scaled weight of indicator $j$ for company $i$.
- $j \in c$ indicates that the summation is over all indicators belonging to component $c$.

The following formula is used to calculate the weights for the components

$$COMP_Weight_{c,i} = \sum_{j \in c} Indicator_Weight_{j,i}$$

Where:

- $COMP_Weight_{c,i}$ is the component weight for component $c$ for company $i$.
- $Indicator_Weight_{j,i}$ is the weight of indicator j for company $i$.
- $j \in c$ indicates that the summation is over all indicators belonging to component $c$.

The definitions of the data points in each component are given below:

## Product Carbon Footprint Practices score

The Product Carbon Footprint Practices score evaluates the company's ability to manage its exposure to risks related to the carbon intensity of its products and its ability to reduce the carbon footprint in its supply chains or in the use of its products and services. This score is equal to the Product Carbon

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Footprint Management score and considers attributes of the company's policies and initiatives. It is expressed as a 0–10 score, where 10 indicates strong initiatives.

**For all industries except the automobile and motorcycle manufacturing industry:**

- **Assessment of energy consumption and/or carbon emissions of suppliers**

**Definition**: Indicates whether the company assesses the carbon emissions and/or energy consumption of its suppliers.

- **Extent of product carbon footprint assessments**

**Definition**: Indicates whether the company assesses the downstream carbon emissions of its products and services.

- **Targets to reduce Scope 3 upstream carbon emissions**

**Definition**: Indicates whether there is evidence the company has quantitative and time-bound targets to reduce Scope 3 upstream carbon emissions. This may include targets regarding the company's supply chain.

- **Energy consumption or carbon emissions assessment of distribution and store operations**

**Definition**: Indicates whether the company assesses the carbon emissions or energy consumption of its distribution centers and store operations.

- **Energy consumption or carbon emissions assessment of manufacturing operations (including suppliers)**

**Definition**: Indicates whether the company assesses the carbon emissions or energy consumption of its manufacturing operations, including suppliers' operations.

- **Energy consumption or carbon emissions assessment of raw materials sourcing**

**Definition**: Indicates whether the company assesses the upstream carbon emissions or energy consumption involved in the production of raw materials.

- **Energy consumption or carbon emissions assessment of transportation and logistics**

**Definition**: Indicates whether the company assesses the carbon emissions or energy consumption of its transportation and logistics operations.

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- **Extent of energy consumption or carbon emissions reduction programs in distribution centers or store operations**

**Definition**: The extent of the company's carbon emissions or energy reduction programs in its distribution centers or store operations. Best practice includes programs in all distribution centers or store operations.

- **Extent of energy consumption or carbon emissions reduction programs in manufacturing operations (including suppliers)**

**Definition**: The extent of the company's carbon emissions or energy reduction programs in its manufacturing operations, including suppliers' operations. Best practice includes programs covering all core production facilities.

- **Extent of energy consumption or carbon emissions reduction programs in raw materials sourcing**

**Definition**: The extent of the company's carbon emissions or energy reduction programs in the sourcing of raw materials, including those sourced by suppliers. Best practice includes programs covering all core products.

- **Extent of energy consumption or carbon emissions reduction programs in transportation and logistics**

**Definition**: The extent of the company's carbon emissions or energy reduction programs in the transportation and logistics stages. Best practice includes improvements in fleet efficiency combined with improvements in load/packaging optimization.

Sources: Company disclosures, CDP, International Council on Clean Transportation, Environmental Protection Agency, European Environment Agency, Korea Energy Industry.

**For the automobile and motorcycle manufacturing industry:**

For companies belonging to the automobile and motorcycle manufacturing industry, the Product Carbon Footprint Management score is calculated using the Product Carbon Footprint Performance score and the Product Carbon Footprint Performance Targets score.

The definitions of the data points in each component are given below:

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## Product Carbon Footprint Performance score

The Product Carbon Footprint Performance score evaluates the company's performance on Product Carbon Footprint metrics, including the company's estimated tailpipe emissions. It is expressed as a 0-10 score, where 10 indicates the highest level of performance.

- **Tailpipe fleet emission performance relative to peers**

**Definition**: Assesses the company's estimated global volume-weighted tailpipe fleet emission performance in comparison to industry peers. Expressed as a 0-10 score, with 10 indicating lowest fleet emissions, relative to peers.

- Product Carbon Footprint industry relative score

  **Definition**: Assesses the company's performance on estimated global volume-weighted tailpipe fleet emission, relative to its peers. Expressed as a 0-10 score, where 10 indicates lowest fleet emissions relative to peers.

- Product Carbon Footprint industry relative trend score

  **Definition**: Assesses the company's performance on the three-year average of year-over-year estimated global volume-weighted tailpipe fleet emission trend, relative to its peers. Expressed as a 0–10 score, where 10 indicates highest decline in fleet emissions, relative to peers.

Sources: Company disclosures; European Environment Agency (EEA), DISCODATA Database; Ministry of Industry and Information Technology (MIIT), Energy and Information Data Center; U.S. Environmental Protection Agency (EPA), Automotive Trends Data

## Product Carbon Footprint Targets score

The Product Carbon Footprint Targets score evaluates the company's initiatives to set, pursue and verify carbon emissions targets. Highest initiatives reflect comprehensive, independently verified targets with strong progress toward achievement. It is expressed as a 0–10 score, where 10 indicates strong initiatives.

- **Carbon emissions target comprehensiveness**

**Definition**: Assesses the extent to which the company has established quantitative climate targets that are comprehensive in scope and complete in coverage. Expressed as a 0-10 score, with 10 indicating comprehensive targets.

- Carbon emissions target footprint coverage (%)

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**Definition**: Assesses the extent to which the company's emission reduction targets cover its total reported emissions. Expressed as a 0–10 score, where 10 indicates that the targets comprehensively cover all relevant emissions.

- Carbon emissions target relevant scope score

**Definition**: Assesses whether the company's emission reduction targets include all financially material scopes. Expressed as a 0–10 score, where 10 indicates full coverage of financially material emission scopes.

- **Carbon emissions target progress**

**Definition**: Assesses the company's progress toward active environmental targets, as well as its historical track record in meeting previously stated targets. Expressed as a 0-10 score, with 10 indicating strong progress towards targets

- Carbon emissions ongoing target progress

**Definition**: Assesses the company's progress toward achieving its emission reduction targets based on current performance. Expressed as a 0–10 score, where 10 all its targets are on track to meet stated targets.

- Carbon emissions target track record

**Definition**: Assesses the company's historical performance in achieving or maintaining sustainability or environmental targets. Expressed as a 0–10 score, where 10 indicates strong performance.

- **Carbon emissions target verification**

**Definition**: Assesses whether the company's disclosed targets are supported by the Science Based Targets initiative (SBTi), where applicable. Expressed as either NULL or 10 score, with 10 indicating presence of verification.

- SBTi long-term target status

**Definition**: Indicates the status of the company's Science Based Targets initiative (SBTi) long-term target, aligned with net-zero or long-term decarbonization goals.

- SBTi near-term target status

**Definition**: Indicates the status of the company's Science Based Targets initiative (SBTi) near-term target, aligned with short- to medium-term emissions reduction goals.

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## Product Carbon Footprint Exposure score

The Product Carbon Footprint Exposure score evaluates the company's exposure to risks on this Key Issue. It is based on the Business Exposure scores only, except for the automobile and motorcycle manufacturing industry. For the automobile and motorcycle manufacturing industry, it is based on the Business Exposure score and the Geographic Exposure score. The Business Exposure score is scored on a 0-10 scale, with 10 corresponding to the highest risk and 0 corresponding to the lowest risk. Refer to the Analyzing risk exposure section of "ESG Ratings Methodology" for additional details. The following formulas are used in the Exposure score calculation:

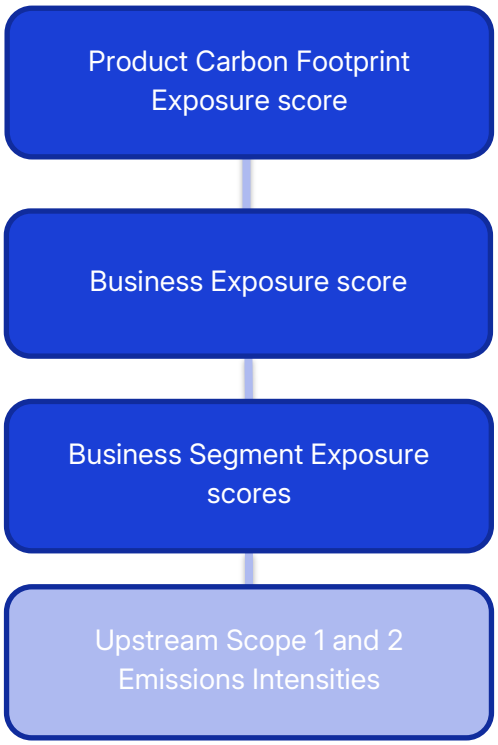
**For all industries except the automobile and motorcycle manufacturing industry:**

$$EXP_{PCF,i} = BUS_{PCF,i}$$

Where:

- $EXP_{PCF,i}$ is the Product Carbon Footprint Exposure score of company $i$.
- $BUS_{PCF,i}$ is the Business Exposure score of company $i$.

**Exhibit 2: Exposure score components for all industries except the automobile and motorcycle manufacturing industry**




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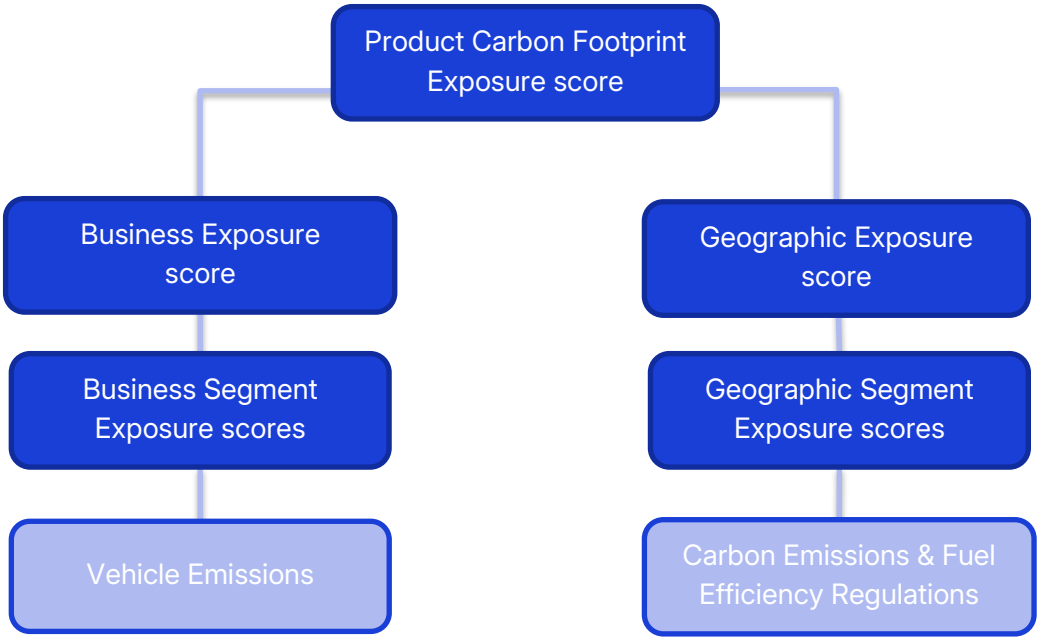
**For the automobile and motorcycle manufacturing industry:**

$$EXP_{PCF,i} = BUS_{PCF,i}(1 + 0.1(GEO_{PCF,i} - 5))$$

Where:

- $EXP_{PCF,i}$ is the Product Carbon Footprint Exposure score of company $i$.
- $BUS_{PCF,i}$ is the Business Exposure score of company $i$.
- $GEO_{PCF,i}$ is the Geographic Exposure score of company $i$.

**Exhibit 3: Exposure score components for the automobile and motorcycle manufacturing industry**




## Business Exposure score

The Business Exposure score is a weighted average of the Business Segment Exposure scores of a company's business segments. For the automobile and motorcycle manufacturing industry, scores are weighted by a company's number of vehicle units sold of a particular vehicle class compared to a company's total number of vehicle units sold, using the following formula to calculate the Business Exposure score:

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**For industries except automobile and motorcycle manufacturing industry:**

$$BUS_{PCF,i} = \sum_{j=1}^{n_i} w_{Revenue,i,j} BSE_{PCF,j}$$

Where:

- $BUS_{PCF,i}$ is the Business Exposure score for company $i$.
- $BSE_{PCF,j}$ is the Business Segment Exposure score for business segment $j$.
- $w_{Revenue,i,j}$ is the weight of business segment $j$ for company $i$ based on the contribution to total company revenue.
- $n_i$ is the number of business segments of company $i$.

**For the automobile and motorcycle manufacturing industry:**

$$BUS_{PCF,i} = \sum_{k=1}^{n_i} w_{Vehicle\ Units,i,k} BSE_{PCF,k}$$

Where:

- $BUS_{PCF,i}$ is the Business Exposure score for company $i$.
- $BSE_{PCF,k}$ is the Business Segment Exposure score for vehicle class segment $k$.
- $w_{Vehicle\ Units,i,k}$ is the weight derived from vehicle class segment $k$ units sold for company $i$ based on the contribution to total vehicle units sold for company $i$.
- $n_i$ is the number of geographic or vehicle class segments for company $i$.

For other industries, scores are weighted by the proportion of a company's total revenue in each business segment, using the following formula to calculate the Business Exposure score:

**Business Segment Exposure scores**

The Business Segment Exposure score is based on a mapping of the business segment as disclosed by the company to a corresponding business activity. MSCI S&C uses the Standard Industrial Classification (SIC) system along with industry-specific adjustments to define business activities. The score associated with a business activity is used to calculate a Business Segment Exposure score.

Each business activity score is determined by the upstream greenhouse gas emissions intensities of the business activity:

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**For industries except automobile and motorcycle manufacturing industry:**

$$BSE_{PCF,j} = GHG_{Upstream.j}$$

Where:

- $BSE_{PCF,j}$ is the Business Segment Exposure score for business activity $j$.
- $GHG_{Upstream.j}$ is the Upstream Greenhouse Gas Emissions score for business activity $j$.

**For the automobile and motorcycle manufacturing industry:**

Each business activity score is determined by a qualitative assessment of specific vehicle class emissions and vehicle weights in a business activity. The highest resulting Business Exposure scores correspond to the passenger vehicle classes with the highest emissions intensities and vehicle weights. Passenger vehicle classes are generally considered to face stricter regulations than commercial vehicles or two-wheelers.

## Geographic Exposure score

The Geographic Exposure score is the weighted average of the Geographic Exposure scores of the countries and regions in which a company operates. Scores are weighted by the proportion of a company's total revenue in each geographic segment using the following equation:

$$GEO_{PCF,i} = \sum_{r=1}^{n_i} w_{Revenue,i,r} GSE_{PCF,r}$$

Where:

- $GEO_{PCF,i}$ is the Geographic Exposure score for company $i$.
- $GSE_{PCF,r}$ is the Geographic Segment Exposure score in region $r$.
- $w_{Revenue,i,r}$ is the weight of total revenue in region $r$ for company $i$.
- $n_i$ is the number of geographic segments for company $i$.

For geographic segments reported as regions (example: Asia Pacific), a nominal GDP-weighted country aggregation is used to calculate region-level scores, using the following equation:

$$GSE_{PCF,r} = \sum_{c=1}^{n_r} w_{GDP,r,c} GSE_{PCF,c}$$

Where:

- $GSE_{PCF,r}$ is the Geographic Segment Exposure score in region $r$.
- $GSE_{PCF,c}$ is the Geographic Segment Exposure score of country $c$.

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- $w_{GDP,r,c}$ is the specific weight of country $c$ within region $r$.

**Geographic Segment Exposure scores**

**For the automobile and motorcycle manufacturing industry:**

The Geographic Segment Exposure score is determined by regulatory requirements for vehicle carbon emissions or fuel efficiency in specific locations. Stricter regulatory requirements for vehicle carbon emissions or fuel efficiency indicates higher Geographic Segment Exposure score.

## Data sources

The following sources are used to determine the Product Carbon Footprint Business Exposure score:

IERS' Comprehensive Environmental Data Archive (CEDA), Refinitiv, MSCI S&C, company disclosures.

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